

Translated

From: e-maxx.ru

Scheduling jobs on one machine

This task is about finding an optimal schedule for n jobs on a single machine, if the job π_i can be processed in t_i time, but for the t_i seconds waiting before processing the job a penalty of $f_i(t)$ has to be paid.

Thus the task asks to find such a permutation of the jobs, so that the total penalty is minimal. If we denote by π the permutation of the jobs (π_1 is the first processed item, π_2 the second, etc.), then the total penalty is equal to:

$$F(\pi) = f_{\pi_1}(0) + f_{\pi_2}(t_{\pi_1}) + f_{\pi_3}(t_{\pi_1} + t_{\pi_2}) + \dots + f_{\pi_n}(\sum_{i=1}^{n-1} t_{\pi_i})$$

Solutions for special cases

Linear penalty functions

First we will solve the problem in the case that all penalty functions $f_i(t)$ are linear, i.e. they have the form $f_i(t) = c_i \cdot t$, where c_i is a non-negative number. Note that these functions don't have a constant term. Otherwise we can sum up all constant term, and resolve the problem without them.

Let us fixate some permutation π , and take an index $i = 1 \dots n-1$. Let the permutation π' be equal to the permutation π with the elements π_i and π_{i+1} switched. Let's see how much the penalty changed.

$$F(\pi') - F(\pi) =$$

It is easy to see that the changes only occur in the i -th and $(i+1)$ -th summands:

$$\begin{aligned} &= c_{\pi_i} \cdot \sum_{k=1}^{i-1} t_{\pi_k} + c_{\pi_{i+1}} \cdot \sum_{k=1}^i t_{\pi_k} - c_{\pi_i} \cdot \sum_{k=1}^{i-1} t_{\pi_k} - c_{\pi_{i+1}} \cdot \sum_{k=1}^{i-1} t_{\pi_k} \\ &= c_{\pi_{i+1}} \cdot \sum_{k=1}^{i-1} t_{\pi_k} + c_{\pi_i} \cdot \sum_{k=1}^i t_{\pi_k} - c_{\pi_i} \cdot \sum_{k=1}^{i-1} t_{\pi_k} - c_{\pi_{i+1}} \cdot \sum_{k=1}^{i-1} t_{\pi_k} \\ &= c_{\pi_i} \cdot t_{\pi_{i+1}} - c_{\pi_{i+1}} \cdot t_{\pi_i} \end{aligned}$$